System Flow: AI-Powered Shipment Management for Crestex Home Textile

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Overview

The system operates in two distinct phases for every shipment:

Phase A — Descending Approval Chain (Top to Bottom): Order is created, AI builds the plan, it gets approved layer by layer downward until it reaches Unit Managers.

Phase B — Ascending Execution Chain (Bottom to Top): Work begins, updates travel upward, new tasks go up one level for approval before descending again, blockers escalate until resolved.

Nothing in Phase B begins until Phase A is fully complete and approved at every level.

PHASE A: Descending Approval Chain

Step 1 — CEO / Relevant Personnel: Order Initiation

Trigger: A buyer PO arrives or a new shipment needs to be created.

Experience:
The CEO or designated personnel opens the New Shipment screen. They have two options:

Option A — Document Upload:
They upload the buyer's Purchase Order (PDF, Word, Excel). The AI parses the document and auto-fills all relevant fields:

- Buyer name and country
- Product type (e.g. King Size Sheet Set, 300TC, Satin Finish)
- Total quantity
- Color/design specifications
- Ex-factory date
- Packaging and labeling requirements
- Certification requirements (Oeko-Tex, GOTS, etc.)
- Special instructions from buyer

The CEO reviews each auto-filled field, corrects anything misread, adds any fields the document didn't cover, and submits.

Option B — Manual Entry:
Standard form with all the same fields filled manually.

What AI does at this point:

- Validates the ex-factory date against current capacity (flags if it's unrealistic)
- Calculates the effective lead time (e.g. "If lab dip is required, effective production window is 42 days, not 60")
- Assigns a Shipment ID and timestamps the initiation
- Generates a risk summary: lead time risk, capacity risk, any known constraints

Output: Shipment record created. Status: Pending AVP Review.

Step 2 — AVP: Task Plan Review for GM

Trigger: CEO submits the shipment. AVP receives a notification.

What AVP sees:

- Full shipment details as submitted by CEO
- AI-generated Task Plan for the GM level, which includes:
    - Key milestones the GM is accountable for (Pre-production meeting, Fabric readiness, Unit target setting, Daily production oversight, Final inspection coordination, Dispatch sign-off)
    - Timeline for each milestone with start and end dates
    - GM-level KPIs for this shipment (e.g. ensure cutting starts by Day 25, ensure stitching target of 600 sets/day is met, ensure ex-factory is not missed)
    - Risk flags the AI has identified at this level

AVP's Actions:

- Review each task assigned to GM level
- Edit task descriptions, adjust dates, add tasks if needed
- Add compliance-specific requirements or buyer-specific instructions
- Approve the GM-level task plan

What happens on approval:

- AVP approval is logged with timestamp
- System moves to GM
- Status: Pending GM Review

Step 3 — GM: Task Plan Review for All Managers

Trigger: AVP approves. GM receives notification with full shipment details and their own task plan.

What GM sees:

- Shipment overview and AVP-approved milestones
- Their own accountability tasks (as set by AVP)
- AI-generated Task Plans for each Manager under them, broken down by unit group:
  - Manager A (Cutting + Stitching): tasks, daily targets, dates, dependencies
  - Manager B (Embroidery + Quilting + Finishing): tasks, daily targets, dates, dependencies
  - Manager C (QC + Packing): tasks, inspection schedule, packing targets, dispatch checklist
- Dependency map: which unit's output feeds into the next unit's start
- Capacity check: AI flags if any unit is already committed to another active shipment during the overlap period

GM's Actions:

- Review all Manager-level task plans
- Adjust line allocations, daily targets, or dates based on floor knowledge
- Resolve any capacity conflicts (e.g. move a task date, split a target across two lines)
- Add notes for specific Managers
- Approve all Manager task plans collectively

What happens on approval:

- GM approval logged with timestamp
- System simultaneously notifies all Managers under this GM
- Status: Pending Manager Review

Step 4 — Manager: Task Plan Review for Unit Managers

Trigger: GM approves. Each Manager receives notification with their specific task plan and the unit-level plans under them.

What each Manager sees:

- Their own tasks and deadlines (as set by GM)
- AI-generated Task Plans for each Unit Manager / Unit Incharge under them:

- ○ Unit Incharge A (Cutting): daily targets per shift, start date, fabric in-house requirement date, handoff date to Stitching
  - ○ Unit Incharge B (Stitching): daily targets per line, start date, dependency on Cutting output, transfer date to Finishing
  - ○ (and so on for their units)
- Any special instructions passed down from GM or AVP

Manager's Actions:

- Review each Unit Incharge task plan
- Adjust shift targets, machine allocations, or sequences based on unit-level knowledge
- Flag any resource gaps (machine under maintenance, skilled worker shortage)
- Approve all Unit Incharge task plans

What happens on approval:

- Manager approval logged
- Unit Incharge tasks are now finalized and locked in the system
- Once ALL Managers under this GM have approved their respective Unit Incharge plans, the system considers Phase A complete
- Status: Plan Fully Approved — Execution Ready

This is the trigger for Phase B to begin.

PHASE B: Ascending Execution Chain

Step 5 — AI Calling Initiates: Unit Incharge Level

Trigger: Full approval chain complete. System clock reaches the scheduled start date for each unit's first task.

Initial Briefing Call — Unit Incharge:
AI calls each Unit Incharge on their registered number:

"This is the production assistant for Shipment #1042, Buyer ABC, King Sheet Sets. Your unit — Cutting — is scheduled to begin on [date]. Your daily target is 800 pieces per shift. Fabric will be issued to you by [date]. I will call you each morning before your shift starts and mid-shift for a progress check. Do you have any questions or issues to flag before we begin?"

Unit Incharge confirms or raises a pre-start concern. Any concern is:

- Logged in the system

- Immediately escalated one level up (to their Manager) for resolution before work begins

Shift Start Call (daily):
Every morning at shift start time:

- AI confirms today's target
- Asks if materials are ready, machines are operational, workers are present
- Any issue raised is logged and Manager is notified immediately

Mid-Shift Check-In:
AI calls mid-shift:

- Requests current piece count
- Calculates whether target is achievable by shift end
- If significantly behind, flags to Manager immediately with projected shortfall

Shift End Confirmation:
AI calls at shift end:

- Final piece count confirmed
- Any issues or observations from the Unit Incharge logged
- Variance from target calculated and recorded
- Summary automatically sent upward to Manager

Step 6 — Updates Ascending: Manager Level

What Manager receives:

- End-of-shift summary from AI for each unit under them (no call needed, dashboard + notification)
- Immediate call if any unit flags an issue during the shift
- Daily consolidated report of all units under them: planned vs actual, issues raised, resolutions

Manager's role in execution:

- Receives flags from AI when units are behind or blocked
- Makes resolution decisions on the call with AI:
    - Authorize overtime
    - Reallocate workers between lines
    - Request material urgently from stores
    - Escalate to GM if beyond their authority
- Any new task created as a result of a resolution (e.g. rework task, expedited material request) is flagged upward to GM for approval before AI starts managing it

Daily upward report:
At end of day, the AI compiles each Manager's unit data and sends a consolidated update to the GM automatically.

Step 7 — Updates Ascending: GM Level

What GM receives:

- Morning briefing call from AI: overall shipment health, which units are on track, which are flagged, what decisions are needed today
- Real-time escalations from any Manager who cannot resolve an issue at their level
- End-of-day consolidated report across all units and all Managers for this shipment

GM's role in execution:

- Reviews the daily briefing and makes cross-unit decisions
- Handles escalations from Managers: reallocates resources between unit groups, adjusts timelines, coordinates with upstream divisions (e.g. requests expedited fabric from Processing)
- Any new task or plan adjustment created at GM level goes up one step to AVP for approval before descending

Upward reporting:
GM-level summary is compiled by AI and sent to Manager/Admin/Compliance and AVP daily.

Step 8 — New Task Creation: Always Goes Up One Level First

Rule: No new task enters the execution chain without approval from one level above where it originates.

How it works:

If a Unit Incharge flags a machine breakdown and a maintenance task needs to be created → AI drafts the task and sends it to the Manager for approval → Manager approves → AI assigns to maintenance team and begins managing it.

If a Manager identifies that a rework batch needs a separate stitching run → AI drafts the rework task plan and sends to GM for approval → GM approves → AI assigns to Unit Incharge and begins calling.

If a GM decides a second shift needs to be added to recover a delay → AI drafts the additional shift task plan and sends to AVP/Manager Admin for approval → Approved → AI begins managing the new shift.

This ensures no unauthorized work begins, every task is traceable to an approval, and the hierarchy is never bypassed.

Step 9 — Blocker Resolution and Escalation

Escalation ladder:

Level 1 — Unit Incharge flags a blocker to AI on a call.
AI immediately notifies the Manager with full context and a resolution window (e.g. 2 hours).

Level 2 — If Manager does not resolve within the window, or the issue is beyond Manager authority, AI escalates to GM with full context including what the Manager said and what action was or wasn't taken.

Level 3 — If GM does not resolve within their window, or the issue requires resource allocation beyond their authority, AI escalates to Manager/Admin/Compliance and AVP.

Level 4 — If unresolved at AVP level within a critical window (e.g. ex-factory date now at risk), AI calls CEO directly with a brief, factual summary and requests a decision.

At every escalation level:

- The person being escalated to receives a call from AI, not just a notification
- They are given full context: what the issue is, when it was first flagged, what has been tried, what the downstream impact is if unresolved
- Their response is logged
- Once resolved, AI notifies all relevant levels downward and resumes normal execution management

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Step 10 — Cross-Department and Cross-Unit Coordination

When one unit's output feeds another, AI manages the handoff:

- 2 days before the scheduled handoff date, AI calls the receiving Unit Incharge to confirm readiness
- On handoff day, AI calls the sending unit to confirm output quantity and quality clearance, then calls the receiving unit to confirm receipt

- Any discrepancy in handoff quantity is immediately flagged to the shared Manager or GM

When coordination is needed between departments (e.g. Home Textile needs fabric from Processing division, or needs QC team from a separate department), AI initiates a coordination call between the relevant contacts, logs the agreement, and follows up on the committed delivery.

What the Completed System Looks Like: Summary

When fully implemented and running on a live shipment at Crestex, here is what the experience looks like from the outside:

A buyer PO arrives on a Monday morning. The CEO or designated person uploads it to the system. Within minutes, all fields are filled, a risk assessment is visible, and the shipment is submitted. By Tuesday, AVP has reviewed and approved the GM task plan. By Wednesday, the GM has reviewed and approved all Manager task plans. By Thursday, all Managers have approved their Unit Incharge task plans. The full approval chain took 3 days instead of the usual week of meetings and WhatsApp threads.

On Day 25, the system clock triggers. The AI calls the Cutting Unit Incharge at 7am before his shift. Every day from that point, the AI is making calls, collecting numbers, logging updates, and surfacing exceptions — without anyone asking it to. The Manager of Cutting and Stitching checks his dashboard at 9am and sees both units are on track. At 11am the Cutting Incharge reports a machine issue on a mid-shift call. The AI calls the Manager immediately. The Manager authorizes overtime on another line. A new task is drafted by AI, sent to the GM for approval, approved in 10 minutes, and assigned — all before lunch.

At 5pm, the GM receives a daily briefing call. Three units are on track, one is 5% behind, one blocker was resolved. The GM makes one decision on the call and hangs up. The Manager/Admin/Compliance opens their dashboard and sees the same information already compiled into a report ready to share with AVP.

On Day 46, QC inspection happens. The AI had already reminded the QC Manager 3 days prior. Results are logged. 200 pieces are flagged for rework. AI automatically creates a rework task, sends it to the Manager for approval, gets it approved, assigns it to Stitching Unit Incharge, and begins following up on it separately from the main production flow — without anyone manually creating a single task.

On Day 55, packing is complete. The AI generates the packing summary. The document checklist is 90% complete — two items are pending. AI calls the responsible persons. By Day 58, everything is ready. The shipment is dispatched. The system closes the shipment and generates a full audit trail: every task, every call, every update, every flag, every escalation, every approval — timestamped and logged.

The CEO, who approved the order on Day 1 and checked the dashboard twice a week, never made a single follow-up call. The GM made one decision per day on average. The Manager/Admin/Compliance reviewed exceptions instead of chasing updates. The Unit Incharges knew exactly what was expected of them every single day. And the shipment went out on time.

Goalchaser Flow for Crestex AI-Calling and Management System